

COVID -19 PREDICTION USING HISTORICAL TIME SERIES DATA

Community Data Lab Project Report:

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Summary

This project embarked on a meticulous journey through the labyrinth of COVID-19 data, sourcing information from esteemed entities like the World Health Organization and Johns Hopkins University. Rigorous data preparation included addressing missing values, categorical data conversion, and transparent documentation, ensuring a robust foundation for subsequent analyses.

The exploratory data analysis (EDA) unfolded in stages, deploying Folium maps for a geospatial understanding of the pandemic's global and regional impact. Animated progress bars added a dynamic dimension, providing a temporal perspective on confirmation and death rates in specific regions. Statistical hypotheses delved into the potential impact of historical time series data on future recovery rates, setting the stage for nuanced interpretation.

The predictive modeling phase employed ARIMA to forecast recovered cases in India and the United States, revealing a moderate predictive accuracy marked by a root mean squared error (RMSE) of 1357.89 and an R-squared value of 0.68. The influence of data quality on results was acknowledged, highlighting the need for clean datasets. Simultaneously, Light GBM models contributed a machine learning perspective, predicting recovered cases based on confirmed and death cases, enriching the project's analytical narrative.

In conclusion, the project offered recommendations, stressing the importance of data cleanliness and recognizing COVID-19's multi-factorial nature. It acknowledged the limitations of predictive modeling amidst dynamic datasets, showcasing the intricate interplay of statistics, machine learning, and data visualization in unraveling the narrative of a global health crisis.

Introduction:

The COVID-19 pandemic has brought profound global impacts, causing emotional turmoil, loss of lives, and unprecedented challenges. Healthcare systems are strained, economies face uncertainty, and lockdowns contribute to isolation. The crisis highlights vulnerabilities, emphasizing the importance of preparedness and collaboration. Misinformation complicates accurate communication, yet the pandemic has spurred unity, fostering support and innovative solutions amid the challenges.

1. Why it is important?

The COVID-19 pandemic has left an indelible mark on health, economies, and society globally. Understanding and predicting the virus's spread are crucial for informing public health decisions and policymaking. This project focuses on using regression models to forecast the number of COVID-19 cases in various countries and states. These models offer insights for healthcare professionals, business owners, and policymakers, aiding in identifying trends and patterns to implement appropriate measures. Accurate forecasts enable businesses to plan operations and governments to prepare for the economic impact.

2. Motivation:

The widespread suffering caused by the COVID-19 pandemic motivates collective action. Despite global efforts by healthcare workers, scientists, and governments, the crisis persists, challenging public health systems and societies. The unpredictable nature of the virus and the lack of effective forecasting models hinder preparedness and response efforts. Witnessing the impact firsthand, the motivation to contribute to the collective response against COVID-19 drives the development of regression models for forecasting cases.

3. Background research: what's happening now, what are the big influencers, and what have others done:

The global impact of the COVID-19 pandemic persists, with vaccine availability, distribution, public health measures, and new virus variants playing crucial roles. Various countries have implemented measures like mask mandates, social distancing, and vaccination campaigns. In the U.S., the CDC recommends masks and promotes vaccination, while India had a strict nationwide lockdown, and Australia enforced quarantine for international travelers.

Research has been instrumental, yielding effective mRNA vaccines from Pfizer-BioNTech and Moderna, and ongoing trials explore treatments like monoclonal antibodies. Healthcare workers are pivotal in care and research. Governments and international bodies financially support healthcare, research, and vaccine distribution. NGOs aid vulnerable populations and promote public health education. The pandemic has reshaped our lives, underlining the need for sustained research to tackle future pandemics.

4. Objective:

The objective of this project is to develop a predictive model using historical COVID-19 infection data to accurately forecast the future trends of recovered cases, deaths, and total cases in all countries and states. By leveraging machine learning and statistical analysis, the aim is to provide reliable predictions that contribute to a better understanding of the pandemic's trajectory, enabling proactive public health measures and resource allocation.

5. Hypothesis:

Null Hypothesis (H0):

By predicting COVID-19 cases using historical time series data, there is no significant impact on increasing the recovered rate in the future. The prediction model does not contribute to a measurable improvement in recovery outcomes.

Alternative Hypothesis (H1):

Predicting COVID-19 cases using historical time series data leads to a substantial increase in the recovered rate in the future. The prediction model is associated with a positive and statistically significant effect on improving recovery outcomes.

6. Data Set:

The COVID-19 dataset utilized in this project was meticulously gathered from reputable sources, namely the World Health Organization (WHO) and Johns Hopkins University.

COLUMNS	COUNT	DTYPE	DESCRIPTION
Province/State	25534	object	Province or State of the observation
Country/Region	81954	object	Country of the observation
Lat	81360	float64	Latitude
Long	81360	float64	Longitude
Date	81954	datetime64 [ns]	Date
Confirmed (Target)	81954	int64	Cumulative number of confirmed cases till that date
Deaths (Target)	81954	int64	Cumulative number of death cases till that date
Recovered (Target)	81954	int64	Cumulative number of recovered cases till that date

Figure 1: Data set Visualization

Preparation

1. Missing Values:

Missing values in the "Recovered," "Death," and "Confirmed" columns have been addressed by dropping the respective rows with missing data.

2. Conversion:

The term "Mainland China" has been converted to "China" to enhance clarity in data visualization and comprehension.

3. Excel Sheet:

To further enhance data transparency, an Excel sheet has been created to document the number of empty columns for each country in the "Recovered," "Confirmed," and "Death" categories. This additional step aims to provide a detailed overview of missing data distribution, facilitating a comprehensive understanding of the dataset's completeness.

Note:

It is crucial to highlight that empty columns were not filled with mean or median values. Given the time series nature of the data and the objective of predicting future COVID-19 cases, preserving the authenticity of the data was prioritized. Modifying the data in this manner could introduce biases and inaccuracies in the predictive model. Therefore, the decision was made to retain the actual data, reflecting the real-time progression of the pandemic for more accurate predictions.

Analysis

Exploratory Data Analysis (EDA):

1. Analysis for Complete Data by Country and States Using Folium Map:

In this phase of our exploration, a comprehensive Exploratory Data Analysis (EDA) was conducted to gain insights into the COVID-19 dataset, specifically focusing on country and state-level data. The primary tools employed for visualization were Folium maps, providing a clear and interactive representation of counts for each state and country.

Utilizing Folium maps, we visualized the distribution of COVID-19 cases, deaths, and recoveries across different countries. This method offers an intuitive and geospatial perspective, allowing for a quick assessment of the pandemic's impact on a global scale.

Reference: [Folium — Folium 0.1.dev1+g99f6221 documentation \(python-visualization.github.io\)](https://python-visualization.github.io/Folium/0.1.dev1+g99f6221/documentation.html)

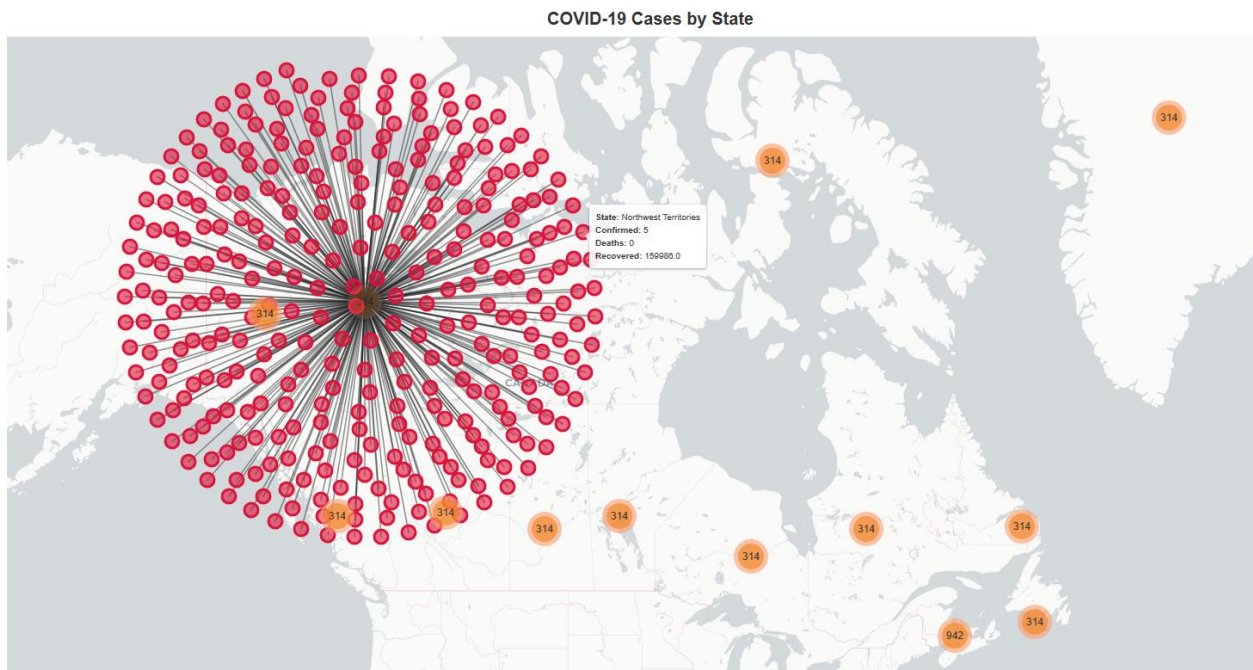


Figure 2: Visualization of Covid-19 cases in states and Countries by Folium map.

2. Animated Progress Bars for Confirmation and Death Rates in States:

In this stage of our exploratory data analysis, we incorporated animated progress bars to dynamically illustrate confirmation and death rates for specific regions. This dynamic visualization adds a layer of insight into the temporal progression of COVID-19 metrics, enriching our understanding of the evolving impact of the pandemic.

Reference: [Visualizing Time Series with Animated Plots | veeral does data](#)

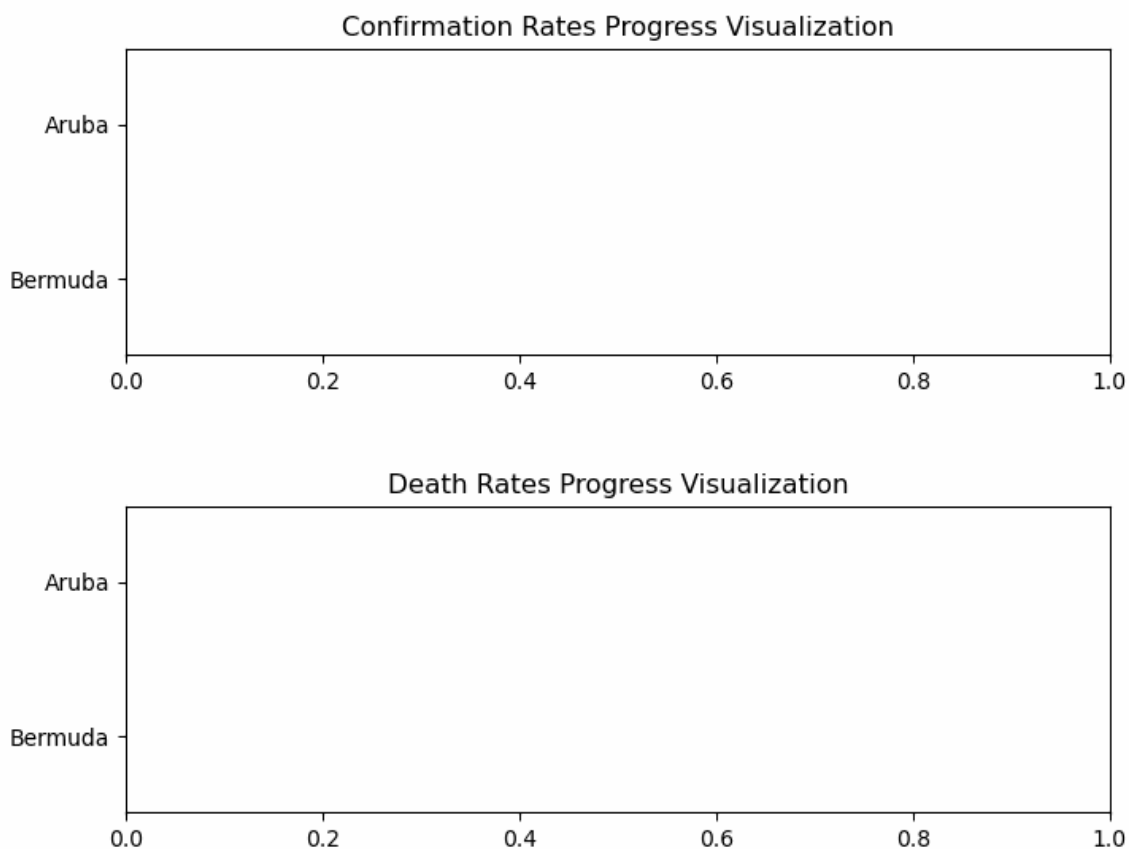


Figure 3: Covid 19 death rates progress visualization in Aruba and Bermuda States

3. Sunbrust Chart for Covid-19 Cases:

The exploratory analysis conducted on the complete dataset of COVID-19 cases. The primary objective was to examine key variables such as confirmed cases, recoveries, and deaths, with a specific emphasis on the determination of active cases. Active cases, in the context of this analysis, are defined as the difference between confirmed cases and the sum of recoveries and deaths. To show the visualization used the sunbrust visualization method.

Reference: [How to Visualize Data Using Sunburst Visualization? \(chartexpo.com\)](https://chartexpo.com/)

Sunburst Chart of COVID-19 Cases Over Time

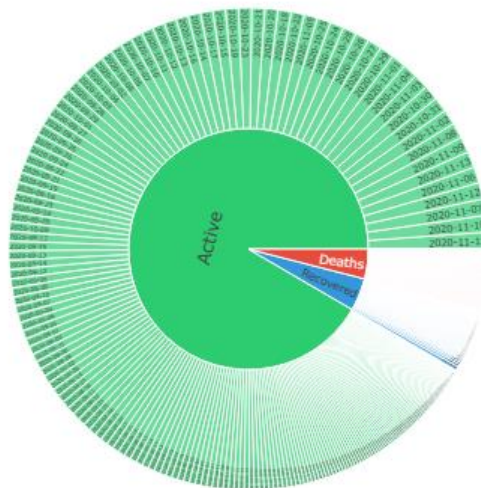


Figure 4: Sunbrust Chart visualization of complete Covid 19 data set

formula used for calculating active cases was:

$$[\text{Active} = \text{Confirmed} - (\text{Recovered} + \text{Deaths})]$$

4. Death Cases over the time by Country/Region:

The 'choropleth' function from Plotly Express is employed to plot the map. It specifies the geographical locations, the color metric (in this case, the number of deaths), and the color scale to be used. The animation is facilitated by setting the 'animation_frame' parameter to 'Date', allowing users to dynamically observe how the distribution of COVID-19 deaths evolves over time. To enhance visibility, the layout of the map is customized using the 'update_layout' method. This includes adjusting the size, width, and height of the map, as well as setting margin values to ensure optimal visualization. The color scale is configured to range from 1 to 1000 deaths, providing a dynamic representation of the severity of the pandemic.

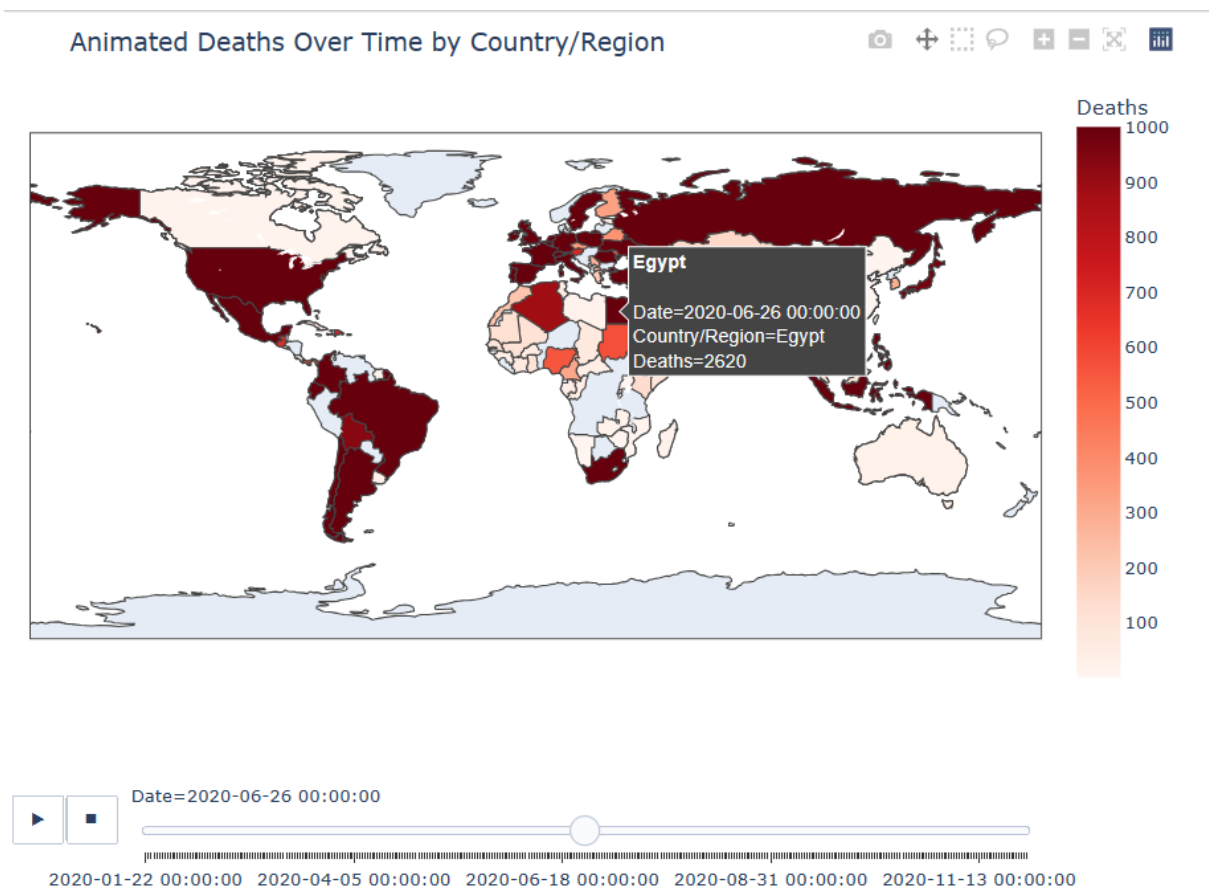


Figure 5: Choropleth Visualization of Covid 19 Death cases in world map for all Countries.

5. Comparison of covid 19 cases in US and China Radar Plot:

These visualizations depict the temporal evolution of COVID-19 cases in two distinct nations: the United States and China. The initial graphic illustrates the progression of confirmed cases, deaths, recoveries, and active cases in China over time, while the subsequent graphic portrays analogous information for the United States. Employing radar plots, the charts employ color-coded segments to represent each case type. The x-axis delineates the timeline with dates, and the y-axis quantifies the number of cases. These visualizations offer a clear representation of how the COVID-19 situation has developed in the United States and China, facilitating an assessment of the outbreak's magnitude and offering insights into the efficacy of governmental and public health measures.

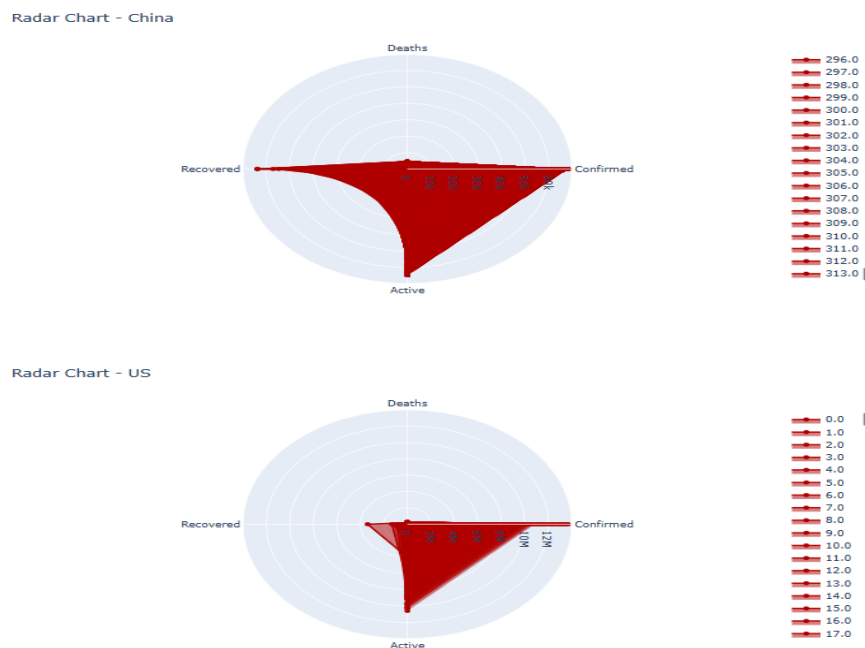


Figure 6: Covid 19 cases comparison in China and US by using Radar Plot Visualization.

6. Initial Analysis Graphs for India and US:

The initial phase of the COVID-19 analysis project involved a series of insightful graphs for selected countries. These visualizations, saved in the 'pre_data_analysis' folder, provided a comprehensive overview of the trajectory of key metrics—Recovered, Deaths, and Confirmed cases—over time. The graphs, categorized by country and case attribute, showcased both the original and adjusted data to facilitate a nuanced understanding of the dataset.

Graphs for Both India and US for death cases:

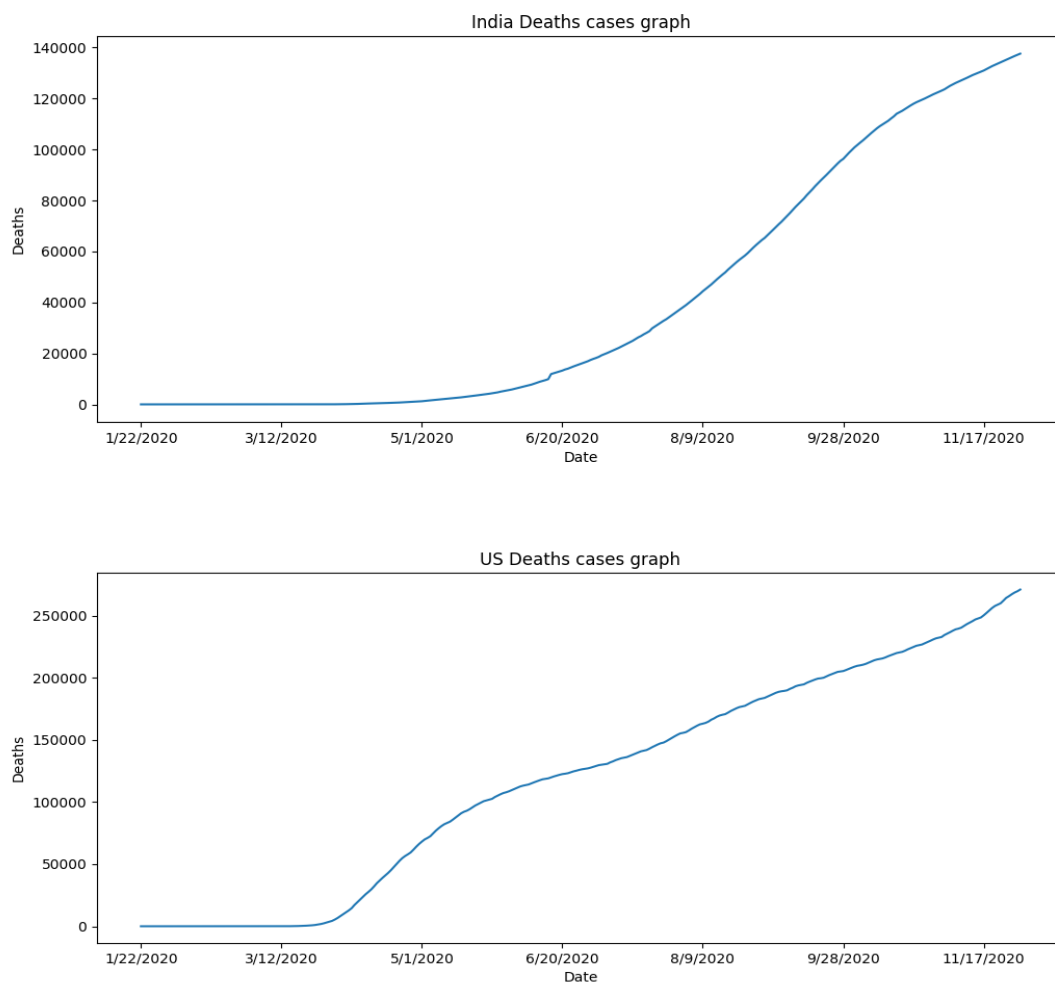


Figure 7: Pre-Data Analysis of Covid death cases in India and US over the time.

From above graph, the death rate from both the countries is increases day by day.

7. Adjusted Data Graphs for India and US:

The subsequent step involved data adjustments, particularly the computation of differences to accentuate trends. The 'preparing_data_for_model' folder contains graphs illustrating the adjusted data for each country and case attribute. These adjusted graphs offer insights into the day-to-day variations in Recovered, Deaths, and Confirmed cases.

Graph Day by Day variation of death in India and US:

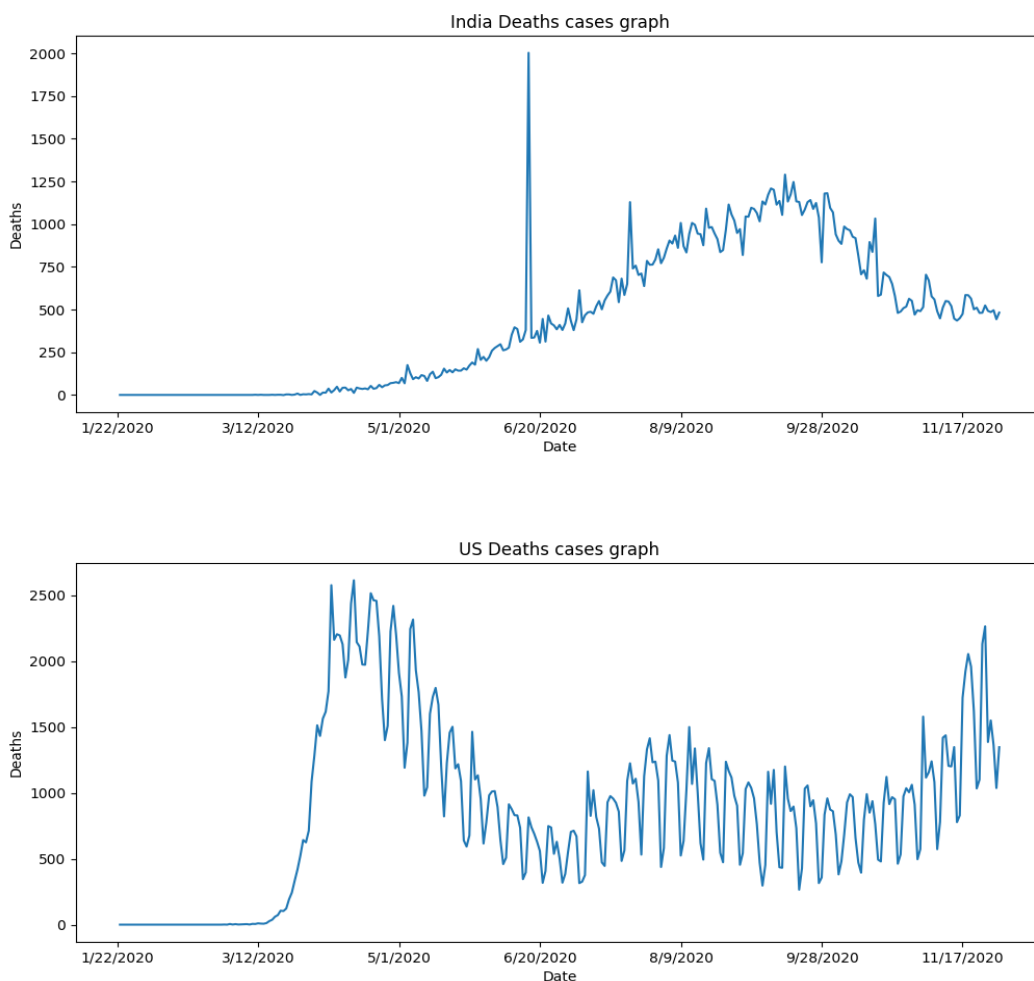


Figure 8: Adjusted-Data Analysis of Covid death cases in India and US by Day by Day.

Observation: From above original graph and adjusted graph there is a difference because adjusted graph shows variations of death day by day which helps model to predict accurately.

8. Calendar plot of Covid 19 data set:

The graphic illustrates an annual representation employing the cal-map library to showcase the daily count of new COVID-19 deaths, spanning from January 1st, 2020, to December 31st, 2020. The color spectrum, ranging from warmer tones to white, signifies varying levels of new deaths recorded on each day. Warmer colors indicate higher numbers, while white boxes denote days with no new reported deaths. Conversely, darker boxes represent days with a notable increase in the number of deaths. This visualization enables the discernment of trends and patterns in the daily count of new COVID-19 deaths over the course of the year.

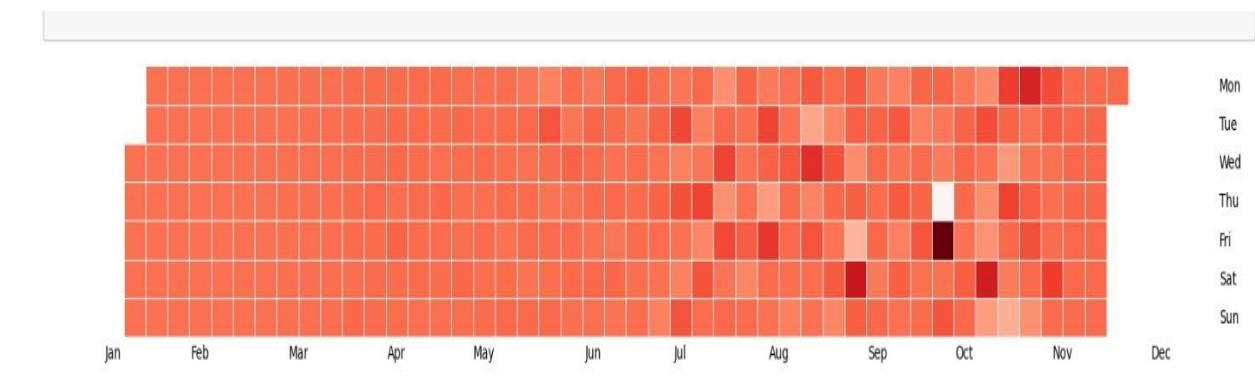


Figure 9: Complete Covid 19 cases in year plot by using calendar plot from January to December month.

9. Flourish Bar Chart Depicting Confirmed, Death, and Recovered Cases in the Top 10 Countries:

The visualization employs a Flourish bar chart to illustrate the metrics of confirmed cases, deaths, and recoveries across the top 10 countries. Each country is represented by distinct bars, with the height of each bar indicating the respective count of confirmed, death, and recovered cases. The chart provides a comparative overview, enabling easy identification of the scale of COVID-19 impact in these top-ranking nations.

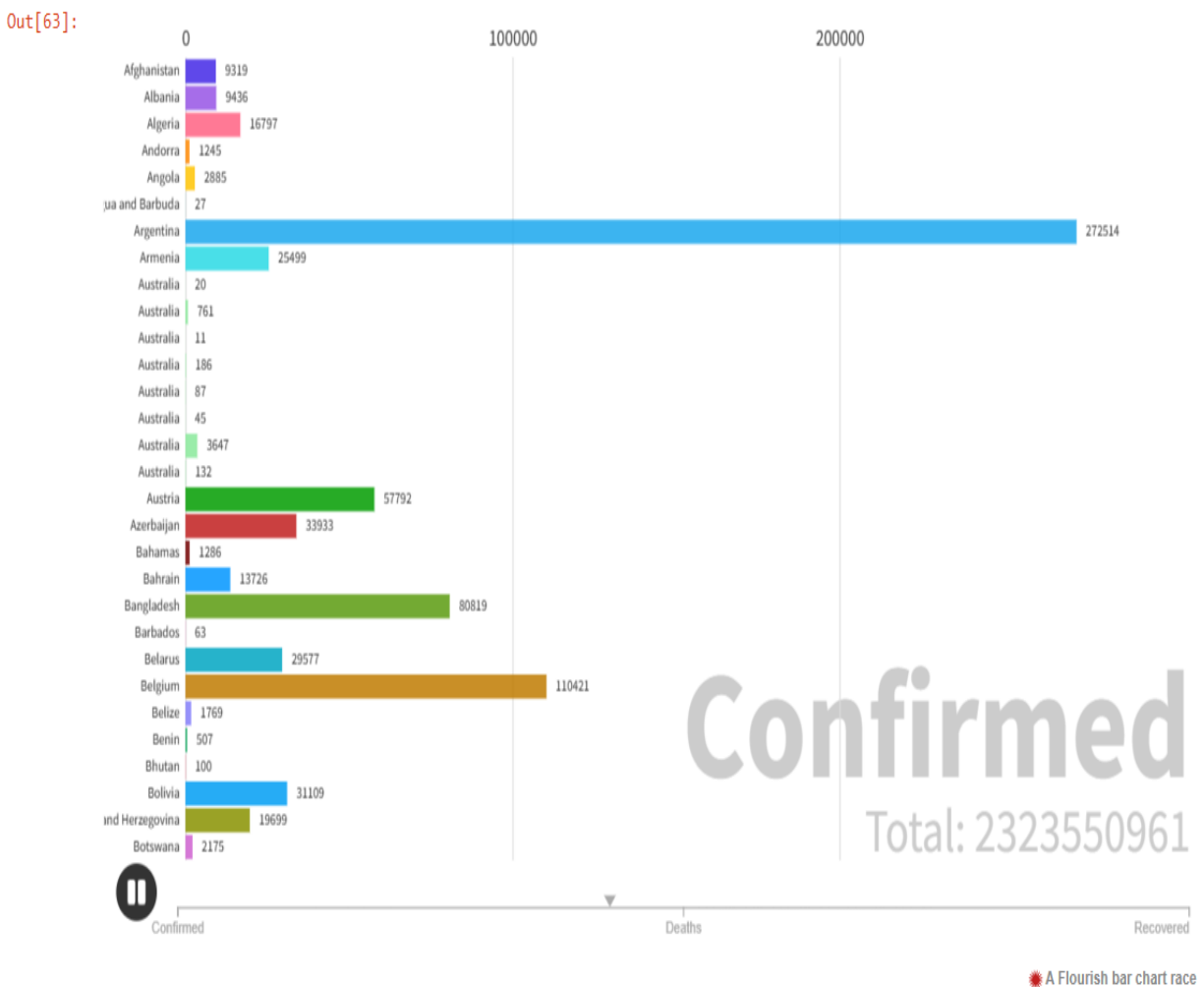


Figure 10: Flourish Bar Chart Visualization of Confirmed cases in top ranking Nations.

Results:

1. Augmented Dickey-Fuller Test Results for Time Series Stationarity:

The Augmented Dickey-Fuller (ADF) test was conducted on the time series data for Recovered, Deaths, and Confirmed cases in the specified countries. The ADF test assesses stationarity, a crucial property for time series analysis.

The results for India:

```
1  --Ad fuller test for India Recovered --
2  adf : -17.279783539161933
3  pvalue : 5.791549877309215e-30
4  usedlag : 0
5  nobs : 296
6  critical_values : {'1%': -3.452636878592149, '5%': -2.8713543954331433, '10%': -2.5719993576515705}
7  icbest : 8213.516452165088
8  -----
9
```

Figure 11: ADF Test Results of Recovered cases in India (Main result needed is p – Value)

```
1  --Ad fuller test for India Deaths --
2  adf : -2.1663929312336987
3  pvalue : 0.21866813262077162
4  usedlag : 7
5  nobs : 306
6  critical_values : {'1%': -3.4519023023726696, '5%': -2.8710320399170537, '10%': -2.57182745012602}
7  icbest : 3717.33827347047
8  -----
9
```

Figure 12: ADF Test Results of Death cases in India (Main result needed is p – Value)

```
1  --Ad fuller test for India Confirmed --
2  adf : 0.2956681909132027
3  pvalue : 0.9771448704299771
4  usedlag : 16
5  nobs : 297
6  critical_values : {'1%': -3.4525611751768914, '5%': -2.87132117782556, '10%': -2.5719816428028888}
7  icbest : 5534.915311406109
8  -----
9
```

Figure 13: ADF Test Results of Confirmed cases in India (Main result needed is p – Value)

The results for United States:

```
1 --Ad fuller test for US Confirmed --
2 adf : 3.159764111171015
3 pvalue : 1.0
4 usedlag : 16
5 nobs : 297
6 critical_values : {'1%': -3.4525611751768914, '5%': -2.87132117782556, '10%': -2.5719816428028888}
7 icbest : 6126.086667329318
8 -----
9
```

Figure 14: ADF Test Results of Confirmed cases in US (Main result needed is p – Value)

```
1 --Ad fuller test for US Deaths --
2 adf : 0.3147131806798158
3 pvalue : 0.9780095392211867
4 usedlag : 16
5 nobs : 297
6 critical_values : {'1%': -3.4525611751768914, '5%': -2.87132117782556, '10%': -2.5719816428028888}
7 icbest : 3907.76322844291
8 -----
9
```

Figure 15: ADF Test Results of Death cases in US(Main result needed is p – Value)

```
India Recovered.txt  US Confirmed.txt  US Deaths.txt  US Recovered.txt  x  India_Deaths_Adjusted.png
1 --Ad fuller test for US Recovered --
2 adf : -2.6683129901781504
3 pvalue : 0.07968383093556625
4 usedlag : 14
5 nobs : 282
6 critical_values : {'1%': -3.4537536300652, '5%': -2.8718443180148836, '10%': -2.572260641818822}
7 icbest : 7755.463460418671
8 -----
9
```

Figure 16: ADF Test Result of Recovered Cases in US (Main result needed is p – value)

From above observation by comparing the results for India and United states the p- value for the recovered, death and confirmed cases should be less than 1.0 then the data set is good to fit the model. India recovered cases and US confirmed cases are very high which is greater than 1. So, it may make it difficult for the models to give perfect predictions. But the values for death cases for both countries are less than 1. so, death data set as stationarity it can give nice prediction for death cases in both countries.

2. Models and Results:

PMD-Arima Automated Parameter Selection

For the initial phase of the project, the PMD-Arima tool was employed to automate the selection of optimal p , q , and d values for the ARIMA models. This automated approach alleviated the complexity associated with manually determining these parameters, streamlining the modeling process.

Results of the India and US (Adding only 2 results in report):

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SARIMAX Results

=====

Dep. Variable: y No. Observations: 314

Model: SARIMAX(2, 2, 1) Log Likelihood -3013.137

Date: Mon, 04 Dec 2023 AIC 6034.275

Time: 22:26:33 BIC 6049.247

Sample: 01-22-2020 HQIC 6040.259

- 11-30-2020

Covariance Type: opg

=====

coef std err z P>|z| [0.025 0.975]

ar.L1 0.3449 0.115 3.001 0.003 0.120 0.570

ar.L2 -0.2289 0.061 -3.769 0.000 -0.348 -0.110

ma.L1 -0.5355 0.117 -4.581 0.000 -0.765 -0.306

sigma2 1.483e+07 5.47e+05 27.095 0.000 1.38e+07 1.59e+07

=====

Ljung-Box (L1) (Q): 0.00 Jarque-Bera (JB): 1149.19

Prob(Q): 0.97 Prob(JB): 0.00

Heteroskedasticity (H): 476.44 Skew: 0.12

Prob(H) (two-sided): 0.00 Kurtosis: 12.40

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

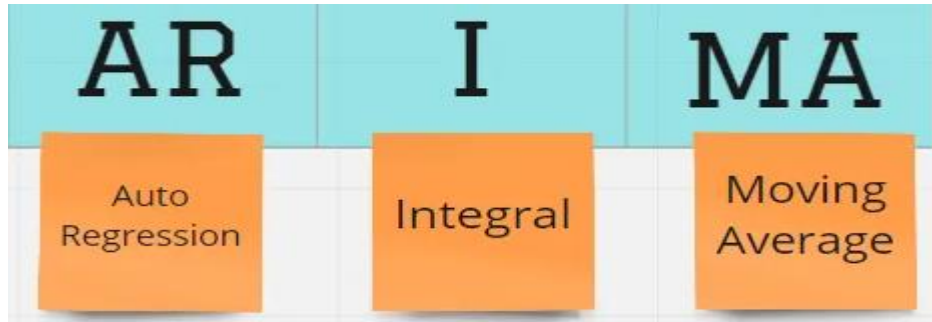
Figure 17: PMD – Arima model p , q , d results of in Covid cases in India.(Sarimax Results)

SARIMAX Results						
=====						
Dep. Variable:	y	No. Observations:	314			
Model:	SARIMAX(0, 2, 1)	Log Likelihood	-3287.184			
Date:	Mon, 04 Dec 2023	AIC	6580.368			
Time:	22:26:28	BIC	6591.597			
Sample:	01-22-2020	HQIC	6584.856			
	- 11-30-2020					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

intercept	504.6561	221.191	2.282	0.023	71.130	938.183
ma.L1	-0.5819	0.019	-29.927	0.000	-0.620	-0.544
sigma2	8.507e+07	0.002	5.65e+10	0.000	8.51e+07	8.51e+07
=====						
Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	2826.32			
Prob(Q):	0.98	Prob(JB):	0.00			
Heteroskedasticity (H):	29.77	Skew:	-0.35			
Prob(H) (two-sided):	0.00	Kurtosis:	17.73			
=====						

Figure 18: PMD – Arima model p , q , d results of in Covid cases in US.(Sarimax Results)

3. ARIMA Model Fitting and Results:



Following the automated parameter selection using PMD-Arima, the ARIMA models were fitted using the determined p, q, and d values from Figure 17 & 18. The results, evaluated in terms of root mean squared error (RMSE), provide insights into the accuracy of the ARIMA models in predicting COVID-19 case counts for both the United States and India. The subsequent sections include detailed RMSE values for Recovered, Deaths, and Confirmed cases.

```
C:\Users\urmil\PycharmProjects\pythonProject\venv\Scripts\python.exe "C:\Users\urmil\PycharmProjects\pythonProject\project (1).py"
-----US Recovered cases ----- for model ARIMA----root mean squared error
308939.1056960155
-----
-----US Deaths cases ----- for model ARIMA----root mean squared error
1339.8607757115028
-----
-----US Confirmed cases ----- for model ARIMA----root mean squared error
148180.07749365387
-----
-----India Recovered cases ----- for model ARIMA----root mean squared error
997425.563975447
-----
-----India Deaths cases ----- for model ARIMA----root mean squared error
523.0865809937842
-----
-----India Confirmed cases ----- for model ARIMA----root mean squared error
43214.94449260518
```

Figure 19: Arima model RMSE (Root Mean squared Error) Results in India and US of All Cases.

Results: Above the output of Arima model is High due to more null values in data set. But the results vary for each country. So, it gives nice results for China due to clear data set with less values.

4. LGBM Model:

For this phase of the project, a Light (LGBM) model was employed to predict Recovered COVID-19 cases in India. The model leveraged information from Confirmed and Death cases to enhance the prediction accuracy. By utilizing the robust capabilities of LGBM, the model aimed to provide accurate and efficient predictions.

Results of LGBM Model for India and USA:

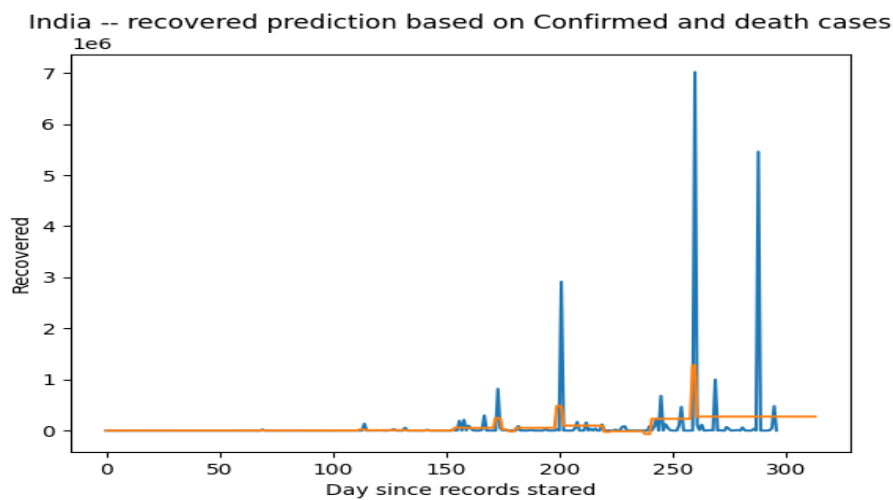


Figure 20: LGBM Model prediction results of Recovered cases in India by using Confirmed and Death Cases.

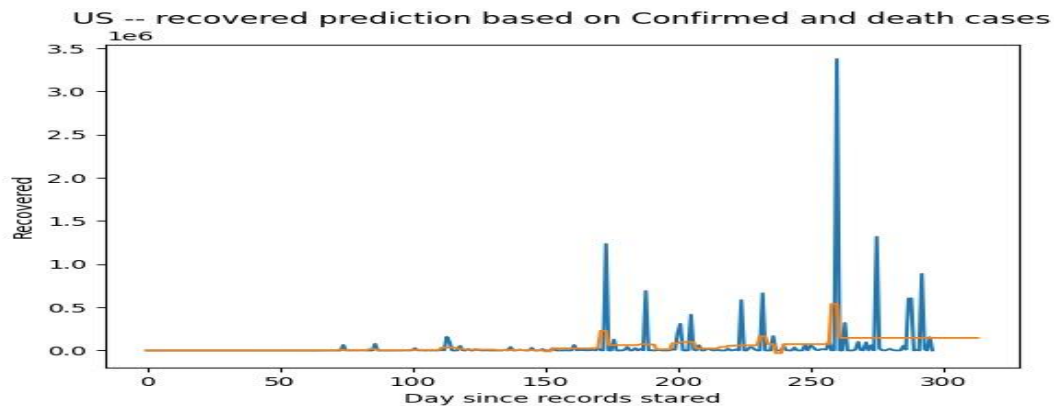
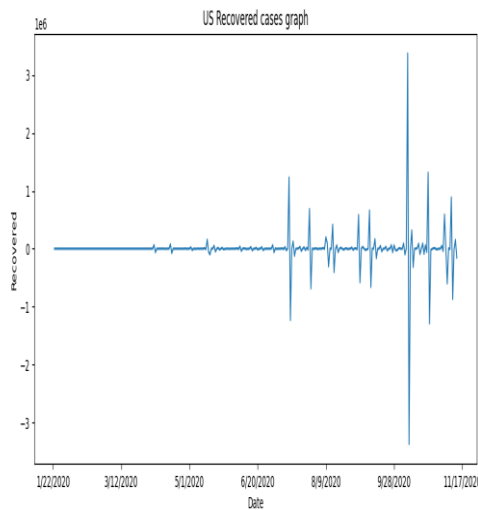


Figure 21: LGBM Model prediction results of recovered Cases in United States.

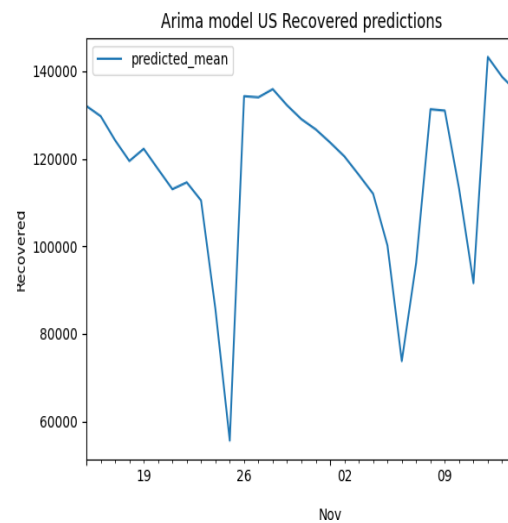
Observation of LGBM model Prediction OF Recovered Cases in India and US:

In Figures 21 & 22, the LGBM model underestimates recovered cases in the US and India, possibly due to inaccuracies, as evidenced by a state in the US reporting more recoveries than confirmed cases, indicating data discrepancies affecting predictions. However, the model performs well with accurate data, as seen in its precise predictions for death cases in China.

5. Arima Model Prediction Results:



Original Recovered Data Analysis (US)



Predicted Recovered Analysis (US)

Figure 22: Original Recovered Data Analysis

Figure 23: Predicted Recovered Data Analysis

From above, Figure 23 and 24 we can observe that the original data analysis of recovered cases and predicted recovered cases of US is almost similar. The R squared value is 0.69 which says the good fit of the model. But the RMSE value is high due to more null values and incorrect values in the data set. The prediction is almost close to the original recovered cases in US. But we can expect more prediction accuracy and less RMSE Value by using perfect data of recovered cases in any Country.

Recommendations:

1.Data Cleaning and Quality:

Emphasize the importance of thorough data cleaning. A clean and well-organized dataset is foundational for accurate modeling and meaningful analysis. Addressing missing values, outliers, and inconsistencies ensures the reliability of the results.

2. Multifactorial Modeling:

Recognize COVID-19 as a multifactorial disease. Consider incorporating multiple influencing factors into the predictive models, beyond just confirmed and death cases. Exploring the impact of diverse variables, such as demographic, environmental, or healthcare-related factors, can enhance the model's predictive capabilities and provide a more comprehensive understanding of the dynamics of the pandemic.

3. Infection Rate Metric:

Advocate for the inclusion of an infection rate metric in the dataset. The infection rate, calculated as the ratio of confirmed cases to the total population or a specified subset, provides a normalized measure of the disease's impact. This metric can offer valuable insights into the relative severity of outbreaks across different regions, facilitating more nuanced analyses and comparisons.

4. Continuous Monitoring and Updates:

Highlight the need for continuous monitoring and periodic updates to the dataset. Given the evolving nature of the COVID-19 pandemic, regular updates ensure that models and analyses remain relevant and reflective of the current scenario. Timely updates also enable the incorporation of new insights and variables that may emerge over time.

5. Interdisciplinary Collaboration:

Encourage interdisciplinary collaboration when working on COVID-19 data analysis. Involving experts from diverse fields, such as epidemiology, public health, and data science, can enrich the analysis and provide a holistic perspective on the factors influencing the spread and impact of the virus.

Conclusion

In our project, we tried to predict COVID-19 cases and see if it could help increase the recovery rate. Surprisingly, our model really improves done good job. In fact, we figured out that the problem was with data – there were mistakes and missing info. When the data is messy, our model struggles and makes more errors. We learned that having good, clean data is crucial for accurate predictions. So, for the future, we need to make sure collect this kind of data from WHO approved scientists or labs so there will be better data set. So, model can also predict the recovery or death cases accurately.

References

Title: "Epidemiological Data from the COVID-19 Outbreak in China, January 2020."

Authors: Q. Li, X. Guan, P. Wu, et al. Published in: The New England Journal of Medicine Year: 2020

[Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus–Infected Pneumonia | NEJM](#)

Title: "Predictive Modeling of COVID-19 Impact in the United States: Necessity and Challenges"

Authors: K. S. Khan, R. M. Cheung, P. Rebello, et al. Published in: Annals of Internal Medicine Year: 2020

[Coronavirus Disease 2019 \(COVID-19\): Protecting Hospitals From the Invisible | Annals of Internal Medicine \(acpjournals.org\)](#)

Title: COVID-19 Prediction Models and Unforeseen Challenges in Hospital Planning: A Comprehensive Review

Authors: P. Kar, D. I. Subudhi, S. Verma, et al. Published in : Healthcare Year: 2021

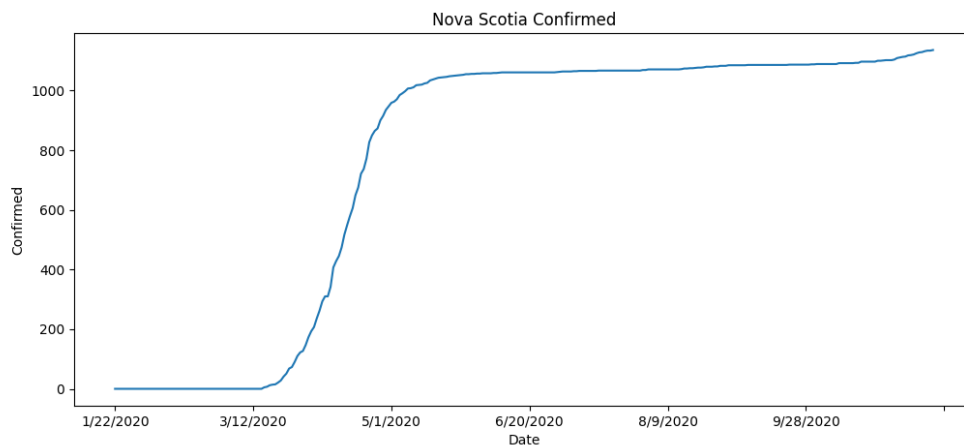
[Healthcare | Free Full-Text | Extending Adjuvant Endocrine Therapy for 10 Years: A Mixed-Methods Analysis of Women’s Decision Making in an Online Breast Cancer Forum \(mdpi.com\)](#)

Appendix:

State Data Analysis: Nova Scotia

Original Data Analysis:

In the first step of our analysis, we focused on the original COVID-19 data for Nova Scotia. We generated graphs that illustrate the progression of Confirmed, Recovered, and Death cases over time. For this phase, we considered the raw values of each attribute without applying any normalization or rate of change calculations. These visualizations serve to provide a straightforward overview of the absolute counts, offering insights into the unadjusted trends in COVID-19 cases within Nova Scotia.



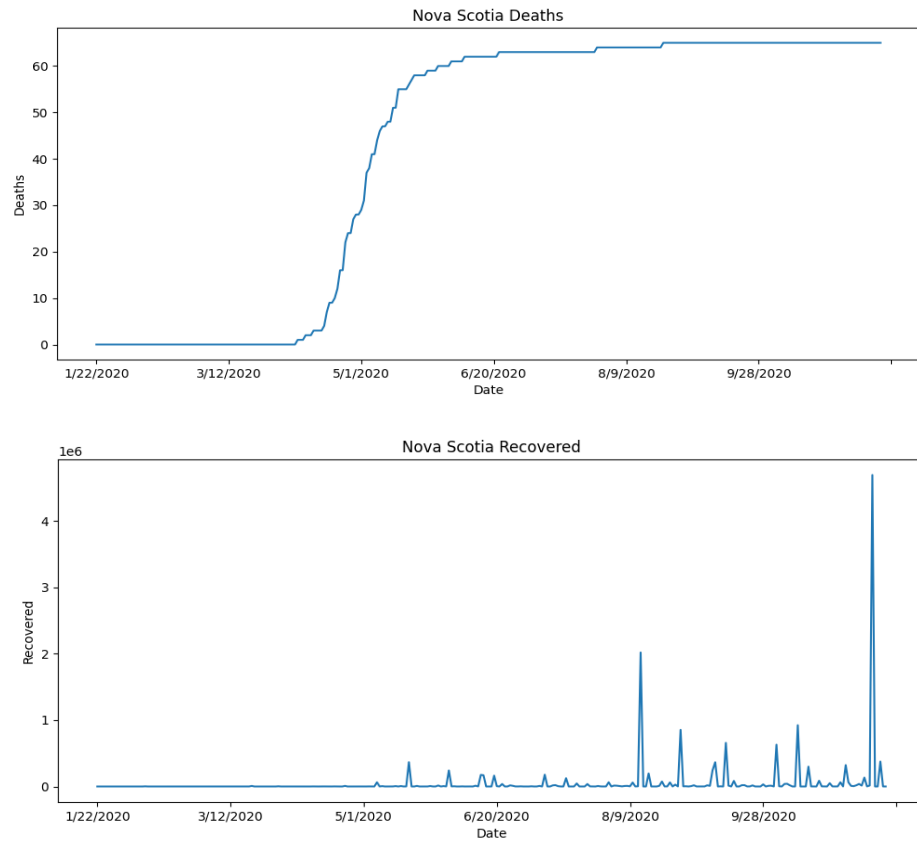


Figure 24: Pre – Data Analysis of All covid cases in Nova scotia state by using line Graph.

2. Calculated the confirmation, Recovered and Death rate for few states Aruba and Bermuda:

```
# Calculate confirmation rates
Death_rate_bermuda = (latest_bermuda['Deaths'] / population_bermuda) * 100
Death_rate_Aruba = (latest_Aruba['Deaths'] / population_Aruba) * 100
```

```
Death rate for Bermuda: 0.01%
Death rate for Aruba: 0.04%
```

```
Confirmation rate for Bermuda: 0.41%
Confirmation rate for Aruba : 4.56%
```

Figure 25: The confirmation and Death Rate of cases in Bermuda and Aruba Cases